M4 terminal, BMS P+
M4 terminal, BMS P−
E−stop: 1 NC out, 2 GND, 3 NC return, 4 GND, 5−6 reset NO, 7 GND, 8 button NO
M4 terminal, HS50 A (+X) +
M4 terminal, HS50 A (+X) −
12 V converter pigtail: 1−2 +VIN, 3−4 −VIN, 5 ON/OFF (low = on), 6 −VIN, 7−9 −VOUT, 10−12 +VOUT
12 V to Jetson (fused, block 10c)
12 V to Pi 5 V buck (fused, block 10c)
12 V to fans (fused, block 10c)
Pi: 3V3, SDA, SCL, GND, E−stop pressed (low)
SPARE_SW: off with the motors
SPARE_ON: always on
M4 terminal, HS50 B (+X) +
M4 terminal, HS50 B (+X) −

1. Battery in and E−stop bank B (LM5069−1, two FETs). Bank A, in series ahead of it, is block 7

The bank is the soft start for all the motor capacitance: BMS sees no inrush. Reset closes it, loop open drops it. No software in the path. Compute is not behind it.
Rs 0.5 mOhm: limit 97−123 A (VCL 48.5−61.5 mV), breaker 160−260 A (VCB 80−130 mV). Above peak draw and JK DOC 80 A. 1.8 W at 60 A.
Start: Cdv 100 nF, gate 10−22 uA, 100−220 V/s, 0.25−0.55 s to full. Bus ceiling 9.4 mF at 2.06 A hot DC SOA line (112 W, 45 K Tj rise). Drive caps unmeasured.
Power limit 652 W at 54.6 V, 480−810 W with tolerance: RPWR = 1.3e5 x Rs x (P − 1.18 mV x VDS / Rs), SNVS452G eq 10, VSNS 6 mV over the 5 mV floor. Timer 4.7 nF: 0.22 ms typ, 0.15−0.39 ms.
Fault check: 810 W is 14.8 A at 54.6 V for up to 0.39 ms. Reviewers read about 16 A at 0.82 ms, 70 C, one FET; TI scales as t^0.346, so about 20.7 A at 0.39 ms: 1.4 x. Read the real curve at layout.
FETs: 0.85 mOhm for the pair, 5.2 W hot at 60 A, 1.3 W at 30 A, the same again in bank A. UVLO low sinks the gate at 2 mA: about 170 us off.
The 2 mA also pulls VBUS down through U1's OUT−GATE zener, against the 54 uA of its internal 1 MOhm SENSE−OUT path, so VBUS reads near 0 V with the banks open.
Body diodes of both banks pass regen back to the pack with the banks open. OVLO tied low: opening on overvoltage would strand regen on the bus.
LM5069−1 latches on a fault. Clear: E−stop cycle, then reset.
BAT spike: the absorber in block 10 (3 x 47 uF) takes the lead current when a bank opens fast, 5.0SMDJ58A is the backstop. FETs and U1 are 100 V.

Block 10. Absorber: a bank trip into a bus short cuts 350−890 A. tools/absorber_sim.py (pack 54.6 V, no pack yet), worst corners: 0.05 Ohm pulse loop (cell AC resistance), 0.5−1 uH, D1 at 71.2 V, C −20 %, trip 8.8−17.5 us. Three 47 uF cans give BAT 80−91 V; one gave 98−118 V. Measure lead L and pulse R at bring−up. 100k bleeds it. BMS close drives about 300 A into three cans for a few us: bench it on the JK.
U2 feed: F1, 22R, ES1D in series, then 1 uF. A shorted U2 puts the pack on VCTL: 22R holds it to about 1.6 A, so SMAJ15A clamps under 19.2 V (Q6 gate 20 V) and F1 (50 A breaking) opens. The series diode keeps C5 from back−driving U2 when BAT collapses, and 22R keeps C39's inrush far under F1's I2t.
Absorber close option (M3), DNP: if the JK trips when it closes onto C38, remove the 0R and fit 2R + the delayed NMOS + its RC: 25 A at close, clamp unchanged (absorber_sim.py).
RUN_A and RUN_B diodes: a U1 pin 2−3 bridge or an EN fault can't drive RUN. FLT reaches UVLO through BAT46W + 10k (2010) with a 12 V zener on FLT_A: FLT low holds UVLO near 1.9 V; a bridged UVLO puts at most 5.5 mA into FLT and 12.7 V on it (15 V recommended).
Converter: F2 20 A on the board end protects the pigtail (U7 already limits at 9.7−12.3 A; F2 backs up a shorted Q7). The DVE has no reverse protection, so the keyed connector carries polarity. U7 PGD high (good) turns on the 2N7002, which pulls ON/OFF to −Vin through 220R: on (0.45 V at 2 mA). 220R limits a pin 5−11 short to 55 mA. 470k and the 12 V zener bias the gate from BRK_IN, so it doesn't depend on VCTL.
1 uF on RST_SRC (M8): a reset held at power−up no longer arms the latch (tools/latch_sim.py, 36 cases). Reset interlock (M1): Q10 shorts RST_SRC until MID is over about 11 V (10 V zener, Q11), and Q14 shorts it while VBUS is over 6−8.5 V (5.1 V zener, hFE spread). Operator rule: wait for VBUS under 3 V. Reset needs MID healthy and the bus down, so a shorted B can't be hard−closed onto a charged bus. After a stop, wait for the drives' caps to bleed. It loads MID by about 30 uA. 33k on U17 VBUS (M13). 22 nF SRC to GND, TI's recommendation for U8. BAT46W (M4). 330 uF on +12V: spare−output shorts and servo back−feed (M9) land in it and the compute loads.

2. Control rail and E−stop latch (no logic IC)

12.1 V, 11 mA. On the battery side, so regen through the body diodes keeps it up with the BMS open. Runs the latch and the clamp. 0.5 W at 54.6 V.
Fed through F1 with its own 1 uF, clamped by a 15 V TVS, reverse diode OUT to IN: block 10.

Seal−in latch, ngspice checked at hFE 80: stays off at power−up even with reset held, reset closes it, a 50 us contact bounce rides through,
loop open drops it in about 100 us, re−closing the loop does nothing, a reset held through a loop cycle does nothing, a new press works 5 ms after release.
RUN 7 V. UVLO and EN_A each take RUN through their own diode (block 10), so a fault on either pin cannot drive RUN. UVLO 3.8 V; needs about 4.8 V on RUN_B rising. Loses VCTL, RUN drops, UVLO drops: bank opens.
Loop 6 mA, wetting for a control−rated NC contact. No 12 V on the connector without a series R. GND between loop out and return, and between return and reset: a short there opens the loop or does nothing.
Brake resistor NC thermostat: wire it in series in the loop harness, no board parts.
Button HB2−BS545, 22 mm mushroom, twist release, 1NC + 1NO, 660 V AC 10 A, control−rated only: NC in this loop, NO to the Pi (block 9). RUN also enables bank A (block 7) and SPARE_SW (block 8).

3. Braking clamp: 56.8 V on, 55.8 V off, 57.5 V worst case, into 2 x HS50 20R in parallel

Two Arcol HS50 20R J in parallel, 10 Ohm total, one on each gusset (+X: J3/J4; −X: J25/J26, block 10c; place all four on the board's top edge, +X half, because the −X leads cross under the board, about 150 mm longer), on the 5 mm countersunk Al plates with thermal compound. Each lead pair is twisted. At the 56.8 V threshold: 5.7 A, 323 W total, 2.84 A and 161 W per resistor. MBRS3100 carries the lead and winding inductance at turn−off, so lead length sets EMI, not FET stress. Each is rated 50 W only on Arcol's 535 cm2 plate, so it works on burst energy: 32 g each, about 29 J/K, so a 3 kJ burst is about +50 C per body. Software caps braking at 50 W average over 60 s and 3 kJ per burst, total. Thermostat 67L120P (opens at 120 C) under one body, in the E−stop loop. Threshold values are rev A's, unchanged.
Self−powered: VCTL lives as long as the bus or the pack does. If the FET sticks on, the resistor's thermostat opens the loop and the bank drops the bus.
TLV1805 drives the FET directly: 100 mA peak, 0.3 us into 4 nF. Rev A's driver, meter, shunt, status FET and NTC are gone, and a smaller FET replaces the IPT015.
Leg stage later: fit 2 Ohm and recheck this FET, 29 A gives about 5 W.

4. Bus capacitors

400 uF, 80 V. The drives have their own. Can is 10 x 16.5: land owed, and it only clears under the higher ceiling, away from the fans.

5. 12 V converter: Murata DVE0125V2N8C on the −X front post plate, pigtail J24 (block 10)

Ahead of the E−stop: motors off, compute stays up. Soft start in block 5a; the converter's negative−logic ON/OFF is held off until U7's PGD is good (block 10).
Off board on the top front rail (frame 3877317), baseplate down, 4x M3 into its threaded baseplate (50.8 x 15.24 mm, 3 mm max depth). J24 sits on the board's top edge, −X half. −Vin and −Vout both go to GND: the robot has one ground, P−.
Pigtail about 150 mm, twisted pairs, 18 AWG: 2 wires each way on the input, 3 each way on +12 V (0.1 V drop at 12 A including contacts). At the converter end: 47 uF 100 V aluminium (its ESR damps the lead, 75−150 nH) plus 1 uF across +VIN/−VIN at the pins. 300 W, 25 A, 94.6 %: about 9 W at the 140 W ceiling, into the plate.
Input UVLO 32−36 V on, 30−34 V off, fixed on the DVE: near an empty pack under sag, compute can drop before the BMS cuts out.
INA228 addresses, no strap on SDA: L arm 0x40, R arm 0x41, aux 0x43, L leg 0x44, R leg 0x45, SPARE_SW 0x47, waist 0x4C, 12 V 0x4D, MID 0x4F. Powered by the Pi's 3.3 V; Pi pull−ups on I2C.

5a. Brick soft start (ahead of the E−stop switch)

Same controller as the bank, so one part family. −2 retries on its own: a compute fault does not need a BMS cycle. UVLO on VIN, OVLO low: on whenever the pack is.
Power−limit start, no dV/dt network: 70 W limit (Rs 5 mOhm, same equation as U1), about 1.3 A at full pack. 4.4 uF plus the brick's own input caps (not published) charges in under 1 ms at 20 uF.
Timer 100 nF: 4.7 ms typ, 3.1−8 ms. Limit 9.7−12.3 A clears the converter's worst draw, about 4.2 A at 36 V. It charges about 48 uF at the converter end plus the DVE's unpublished internal capacitance at the low power limit.
Fault check: 87 W is 1.6 A for up to 3.8 ms, against 12 A at 10 ms on the datasheet SOA. Q7 is the linear FET for that reason.
Compute is on whenever the BMS is on. Loss at 3 A: 0.05 W shunt, 0.03 W FET.

6. Limb outputs: terminal pair, 1 mOhm shunt, INA228. Fuses inline in the pigtails.

1 mOhm: 60 A peak is 60 mV, ADCRANGE 0: [illegible]
Legs DNP: fit the same parts when they arrive.
Returns share two GND studs, three limbs each: check the stud and its lug for the summed current.

7. E−stop bank A: second series bank, own driver, ahead of bank B

Own driver: a FET usually dies shorted gate to drain, which would pull a shared gate net to the pack and hold the other bank on. RUN is the only shared signal, and each bank takes it through its own diode.
Both banks face the same way, drains toward the pack, so regen stays open. Bank B keeps U1, the shunt and the soft start.
Close order: U8 FLT holds U1 UVLO low (about 1.9 V, through BAT46W + 10k) until BST is up (about 57 ms typ, 35−115 ms with 2.2 uF), so A closes first by logic, not by timing. The diode only lets FLT pull down: a failed−high FLT can't enable B. A closes onto MID only. FLT also blocks B if U8 has no charge pump or has tripped.
Off edge: EN falls first, 4.6 us typ, 6.5 us max, then G1PD pulls 390 nC through 22R: about 2−3 us. On a bus short U1's breaker can open B first, so the spike can land on MID with A on: the absorber (block 10) bounds BAT, and so MID, and U17 has its own clamp.
Short−circuit comparator on RS1: 40.2k gives 99 mV, 158−242 A, above U1's limit. CS+/CS− through 0R with a DNP cap: TI's RC placeholder against nuisance trips. TMR 100k only (C29 DNP): fastest response, 4.3 us, still latch−off, cleared by an E−stop cycle. It covers reset with B shorted: A opens, and FLT holds B off.
Driver off holds gate to source with no path to GND, so MID can be biased: 47k + 47k, 0.47 x pack nominal, 0.6 mA, 16 mW each. Hot FET leakage (200 uA per bank at 125 C) moves MID by up to about 4 V. U17 reads MID through 33k with a 68 V zener at the pin (85 V max): calibrate the 3 % gain.
Self−test before reset, pack 36−54.6 V, BMS telemetry for pack voltage: MID 0.33−0.60 x pack and VBUS under 3 V is healthy (U11 reads V_L1: use the highest limb meter). MID within 2 V of pack: A shorted. MID under 3 V: B shorted. VBUS high: both shorted, or the bus has not bled down yet, about 11 s on board caps, longer with drives.
U17 uses VBUS only, IN+ and IN− on GND. VBUS alone cannot see one shorted bank, which is why MID is read.

8. Spare 12 V outputs

SPARE_SW: RUN enables it, so it drops with the motor bank. The divider keeps 7.7 V of RUN under the 6 V EN limit. Limit 3.87−4.42 A (487R). dV/dt 3.3 nF: about 12 V/ms. One switch: a shorted U9 stays on. Current and volts at 0x47, 5 mOhm, ADCRANGE 1.
SPARE_ON: block 8a. Both spares sit behind RS8, so the 12 V meter counts them.

8a. SPARE_ON eFuse

Unswitched, about 2 A limit. Same part as U9: a short stops in 5 us and leaves compute up.

10b. Absorber close option (M3): 0R fitted, the rest DNP

9. E−stop pressed: button NO contact to the Pi

Low means pressed. Pi GPIO input, no Pi pull−up needed. 3.3 V through 3.3k gives the contact 1 mA of wetting.
A harness short to the 54.6 V bus: the BAT46W blocks the pull−up (block 10), and 100k leaves 0.5 mA in the clamp pair. Open contact reads about 3.0 V.
Information only: the NC loop does the stopping.

10. Rev B2 protection parts (refs appended so earlier refs keep their numbers)

10c. Rev B4 review parts

Rev B4 review: two more absorber cans (three in all). U2 feed 22R and series ES1D. FLT diode BAT46W (pull−down only). 68 V zener on U17 VBUS. 220R in Q12's drain. U8 sense RC placeholder: 0R fitted, 1 nF DNP. Schottky on SPARE_ON (and D5 on SPARE_SW). VBUS bleeder footprint, DNP: fit it only if the drives don't bleed the bus. Fuses on the Jetson, Pi buck and fan outputs (the DVE limits at about 30 A). SMBJ15A on +12V: servo back−feed clamp, 16.7 V min breakdown, so it never conducts at 12 V.

Rev B: a part goes in only if leaving it out would hurt. Parts list: hardware/electrical/pdb/REV_B_PAR
marengo
Sheet: /
File: marengo_pdu_revb.kicad_sch
Title: marengo power board rev B
Size: A0 | Date: 2026−11−06 | **Rev: B−wip**
KiCad E.D.A. 10.0.8 | Id: 1/1